

0. 考场使用顺序 Exam Use

1. 画图/写joint: 找 parents, 套 $\prod_i P(X_i \mid \text{Parents}(X_i))$ 。
2. 数参数: Boolean node with k parents needs 2^k params.
3. 独立性: 找 all undirected paths, 再按 chain/fork/collider 判 active。
4. 作答格式: 先 True/False, 再写 active path 或 all paths blocked。

1. Big Picture 大图景

Probabilistic model 概率模型: 用概率描述现实的一部分; 一定是 **simplification**, 目的是由 evidence 推 unknowns。

Bayesian Network / BN: DAG + CPT. 图给直接依赖关系; 每个节点存 $P(X \mid \text{Parents}(X))$ 。

Why BN: Full joint 太大; BN 利用 conditional independence 只存 local CPTs。

A Bayesian network is a directed acyclic graph with one conditional probability table for each variable. It compactly represents the joint distribution by exploiting conditional independence.

2. 核心公式 Must Know

Product rule: $P(x, y) = P(x \mid y)P(y)$ 。

Chain rule: 任意变量顺序都成立:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1})$$

BN factorization 最重要:

$$P(x_1, \dots, x_n) = \prod_i P(x_i \mid \text{Parents}(X_i))$$

Write one factor per node: root nodes use marginals; non-root nodes condition on parents.

Local Markov assumption:

$$X_i \perp\!\!\!\perp \text{non-descendants of } X_i \mid \text{Parents}(X_i)$$

中文: 给定 parents 后, 一个节点与其他非后代变量无关。

3. 参数量 Parameter Counting

Boolean full joint: N 个 Boolean variables:

$$\text{independent numbers} = 2^N - 1$$

因为所有 joint entries 加起来为 1。

Boolean BN node: 若节点 X 有 k 个 Boolean parents, 则 CPT 需要

$$2^k$$

个参数 (有 2^k 行; 每行 $P(T \mid \text{parents}) + P(F \mid \text{parents}) = 1$, 所以 Bernoulli 每行只需写 $P(X=T \mid \text{parents})$) 。

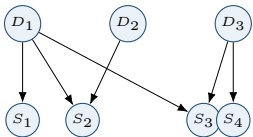
Total BN params: 所有节点相加:

$$\sum_i 2^{|\text{Parents}(X_i)|}$$

英文模板: A full joint over N Boolean variables needs $2^N - 1$ parameters. A BN needs the sum of local CPT sizes, so it can be much smaller when each node has few parents.

4. Tutorial 1: Network Basics

题意: Diseases D_1, D_2, D_3 mutually independent; symptoms caused by: $S_1 \leftarrow D_1, S_2 \leftarrow D_1, D_2, S_3 \leftarrow D_1, D_3, S_4 \leftarrow D_3$ 。



Joint:

$$P(D_1, D_2, D_3, S_1, S_2, S_3, S_4) = P(D_1)P(D_2)P(D_3)P(S_1 \mid D_1)P(S_2 \mid D_1, D_2)P(S_3 \mid D_1, D_3)P(S_4 \mid D_3)$$

Params: $1 + 1 + 1 + 2 + 4 + 4 + 2 = 15$ 。若无 conditional independence / clique by order:

$$1 + 2 + 4 + 8 + 16 + 32 + 64 = 127 = 2^7 - 1。$$

5. Independence 独立

Independence:

$$X \perp\!\!\!\perp Y \iff P(x, y) = P(x)P(y)$$

对所有取值成立。等价: $P(x \mid y) = P(x)$ 。中文: 知道 Y 不改变对 X 的判断。

Conditional independence:

$$X \perp\!\!\!\perp Y \mid Z \iff P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

对所有 $P(z) > 0$ 的取值成立。等价: $P(x \mid y, z) = P(x \mid z)$ 。中文: 知道 Z 之后, Y 不再提供额外信息。

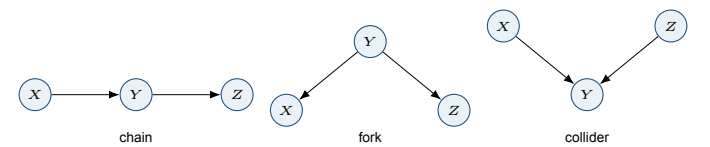
X and Y are conditionally independent given Z iff knowing Y gives no additional information about X once Z is known.

6. 三种结构 Three Patterns

Type	Path	Rule
Chain 链	$X \rightarrow Y \rightarrow Z$	observe $Y \Rightarrow$ block
Fork 叉	$X \leftarrow Y \rightarrow Z$	observe $Y \Rightarrow$ block
Collider 碰撞	$X \rightarrow Y \leftarrow Z$	unobserved $Y \Rightarrow$ block ; observe Y or descendant $\Rightarrow$ open

口诀

- Chain/Fork: **观测中间, 路径堵住。**
- Collider: **不观测中间, 路径堵住; 一观测中间或后代, 路径打开。**



7. D-Separation 判题流程

要判断 $X \perp\!\!\!\perp Y \mid E$:

1. ignore directions: 先把图当无向图, 找 X 到 Y 的所有路径。
2. 每条路径逐个 triple 检查: 只检查路径上的**中间节点**, 不要把 endpoints X, Y 当 blocker。chain/fork 的中间点在 E 中 $\Rightarrow$ path blocked。collider 的中间点或其 descendant 在 E 中 $\Rightarrow$ open; 否则 blocked。
3. 一条路径只要有一个 blocked triple, 整条路径 blocked。
4. All paths blocked $\Rightarrow$ d-separated $\Rightarrow$ independent guaranteed。
5. At least one active path $\Rightarrow$ not d-separated $\Rightarrow$ independence not guaranteed。

标准句: There exists an active path ..., so the variables are not d-separated. Therefore the claimed independence is false.
或: All undirected paths are blocked by the evidence, so the variables are d-separated and conditionally independent.

8. Explaining Away 解释性抵消

结构 $D_1 \rightarrow S_3 \leftarrow D_3$ 是 collider。

- 未观察 S_3 : $D_1 \perp\!\!\!\perp D_3$, 路径 blocked。
- 已知 $S_3=T$: 路径打开, $D_1 \not\perp\!\!\!\perp D_3 \mid S_3$ 。

直觉: 症状已出现, 若知道 D_3 更可能, 就会降低 D_1 的必要性; 两个原因开始互相影响。

9. Tutorial 1 观察题

Observe $S_4 = T$ only: S_4 只直接由 D_3 引起。 D_1 到 S_4 的路径 $D_1 \rightarrow S_3 \leftarrow D_3 \rightarrow S_4$ 在 collider S_3 blocked; D_2 到 S_4 也 blocked。
Learn only about D_3 。

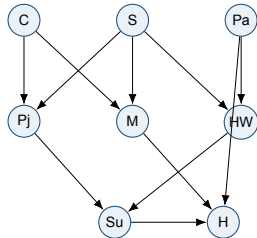
Given $S_3 = T$, observe $S_4 = T$: S_3 被观测, collider $D_1 \rightarrow S_3 \leftarrow D_3$ 打开; S_4 告诉我们 D_3 , 进而也影响 D_1 。
Learn about D_3 and D_1 ; still not D_2 because paths from D_2 remain blocked at S_2 。

10. Tutorial 2: D-Separation Graph

Edges:

$$C \rightarrow Pj, C \rightarrow M, S \rightarrow Pj, S \rightarrow M, S \rightarrow HW, Pa \rightarrow HW, Pa \rightarrow H, Pj \rightarrow Su, HW \rightarrow Su, M \rightarrow H, Su \rightarrow H$$

Legend: C =Creative, S =Smart, Pa =Party, Pj =Project, M =Mac, Su =Success, H =Happy.



11. Tutorial True/False 直接套

Q1 Party independent of Success given HW?

False. Direct path $Pa \rightarrow HW \rightarrow Su$ blocked by observed HW , but

$$Pa \rightarrow HW \leftarrow S \rightarrow Pj \rightarrow Su$$

is active because HW is a collider and is observed.

Q2 Creative independent of Happy given Mac?

False. Path $C \rightarrow M \rightarrow H$ blocked by observed M , but

$$C \rightarrow Pj \rightarrow Su \rightarrow H$$

remains active.

Q3 Party independent of Smart given Success?

False. Path

$$Pa \rightarrow HW \leftarrow S$$

is opened because collider HW has descendant Su in evidence. Also $Pa \rightarrow HW \rightarrow Su \leftarrow Pj \leftarrow S$ can be considered via observed collider Su .

Q4 Party independent of Creative given Happy?

False. Path

$$Pa \rightarrow H \leftarrow M \leftarrow C$$

is opened because collider H is observed.

Q5 Party independent of Creative given Success, Project, Smart?

True. All paths blocked: through H blocked at unobserved collider H ; through HW , S blocked at observed non-collider S ; through HW , Su , Pj blocked at observed non-collider Pj .

12. D-Separation 作答模板

Claim: X independent of Y given E ? True/False

Step 1: List an undirected path $X - \dots - Y$.

Step 2: For each middle node:

classify by arrows ON THIS PATH, not by global intuition

chain/fork + observed $\Rightarrow$ blocked

collider + not observed/no observed desc $\Rightarrow$ blocked

collider + observed/observed desc $\Rightarrow$ open

Step 3:

if one active path exists $\Rightarrow$ False

if all paths blocked $\Rightarrow$ True

Final sentence:

"There exists an active path ...", or

"All paths are blocked by the evidence."

14. 画 BN 题模板 Draw a BN

- 1. 列变量: diseases/symptoms/events 全部变 node。
- 2. 找 root causes: 没有 parents 的变量写在上面。
- 3. 读句子 “A can cause/influence B” $\Rightarrow A \rightarrow B$ 。
- 4. 若题给 variable ordering: 按顺序处理 X_i , parents 只能从 earlier variables 里选; 加到 X_i 与其余 earlier variables 条件独立为止。
- 5. 检查 DAG: 不能有 directed cycle。

Use one node per random variable. Add directed edges from direct influencers/parents to the variable whose CPT depends on them.

15. 写 Joint 题模板

For each node Xi:
if Xi has no parents: write P(Xi)
else: write P(Xi | Parents(Xi))
Multiply all factors once.

例: Alarm Network

$B \rightarrow A \leftarrow E, A \rightarrow J, A \rightarrow M:$

$$P(B, E, A, J, M) = P(B)P(E)P(A \mid B, E)P(J \mid A)P(M \mid A)$$

Specific assignment:

$$P(+b, -e, +a, -j, +m) = P(+b)P(-e)P(+a \mid +b, -e)P(-j \mid +a)P(+m \mid +a)$$

16. 数 CPT 参数题模板

Boolean node: k parents $\Rightarrow 2^k$ rows, 1 free parameter per row.

$$\#params(X) = 2^k$$

Non-Boolean general form: 若 X 有 r 个值, parents combinations 为 q :

$$\#params = (r - 1)q$$

Full joint: variables domain sizes d_i :

$$\#params = \prod_i d_i - 1$$

Tutorial medical count

CPT	k	params
$P(D_1), P(D_2), P(D_3)$	0	$1 + 1 + 1$
$P(S_1 \mid D_1)$	1	2
$P(S_2 \mid D_1, D_2)$	2	4
$P(S_3 \mid D_1, D_3)$	2	4
$P(S_4 \mid D_3)$	1	2
Total		15

17. “为什么更紧凑” 答案

A full joint table over 7 Boolean variables needs $2^7 - 1 = 127$ independent numbers. The BN only needs 15 numbers because each symptom is conditionally independent of many other variables given its direct disease parents. Thus the BN stores local CPTs instead of the full joint.

数参数步骤: 先看每个 node 有几个 parents; Boolean node 写 2^k ; root 是 $2^0 = 1$; 最后把所有 CPT 加总。若题说 “no conditional independence”, 按 ordering 让后面变量依赖所有前面变量, 所以是 $1 + 2 + 4 + \dots + 2^{N-1}$ 。

18. Independence 公式判断题

如果题给概率表

直接代公式:

$$X \perp\!\!\!\perp Y \iff P(X, Y) = P(X)P(Y)$$

$$X \perp\!\!\!\perp Y \mid Z \iff P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$$

或:

$$P(X \mid Y, Z) = P(X \mid Z)$$

如果题给 BN 图

不要代数值; 用 d-separation. 图上 blocked 才是 graph-guaranteed independence。

19. Active Path 速查表

Triple	Middle in evidence?	Active?
chain: $A \rightarrow B \rightarrow C$ or $A \leftarrow B \leftarrow C$	No	Yes
chain: $A \rightarrow B \rightarrow C$ or $A \leftarrow B \leftarrow C$	Yes	No
fork: $A \leftarrow B \rightarrow C$	No	Yes
fork: $A \leftarrow B \rightarrow C$	Yes	No
$A \rightarrow B \leftarrow C$	No + no observed desc	No
$A \rightarrow B \leftarrow C$	B or desc observed	Yes

一条 path active 当且仅当路径上 每一个 triple 都 active; 任何一个 triple blocked, 整条 path blocked。

20. 常见图的独立性

Toothache-Cavity-Catch: $T \leftarrow C \rightarrow Ca$:

$$P(T, C, Ca) = P(C)P(T \mid C)P(Ca \mid C)$$

$$T \perp\!\!\!\perp Ca \mid C$$

中文: 给定是否蛀牙后, 牙疼和探针勾到互不再提供信息。

Alarm: $B \rightarrow A \leftarrow E, A \rightarrow J, M:$

$$J \perp\!\!\!\perp M \mid A$$

但是 $B \not\perp\!\!\!\perp E \mid A$, 因为观察 alarm 打开 collider (explaining away) 。

21. Explaining Away 英文句

In a collider structure $X \rightarrow Z \leftarrow Y$, X and Y are marginally independent because the path is blocked at Z . Once Z or a descendant of Z is observed, the collider is opened, so information about one cause changes the probability of the other cause. This is called explaining away.

22. 信息更新题怎么写

题型: “What variables do we gain information about after observing E ?”

- 1. 对每个 candidate variable X , 问 X 与 evidence E 是否 d-separated。
- 2. 若 not d-separated, 有 active path $\Rightarrow$ observing E can give information about X 。
- 3. 若 d-separated, all paths blocked $\Rightarrow$ no information about X guaranteed by graph。

Observe S4:
D3 – S4 active => learn D3.
D1 – S3 – D3 – S4 blocked at S3 collider
D2 paths blocked at S2/S3
=> only D3.

Given S3, observe S4:
S3 observed opens D1 -> S3 <- D3
=> learn D3 and D1, not D2.

标路径小技巧: 在每条 path 的中间点上写 O/X。普通点 observed = X; collider unobserved = X; collider observed/desc observed = O。整条 path 有任一 X 就 blocked; 全 O 才 active。

23. 中英词典 Mini Dictionary

English	中文 / 考试含义
random variable	随机变量, 一个 node
evidence	已观测变量, conditioning set
query variable	想推理/判断的变量
parent / child	父节点 / 子节点
descendant	后代, 沿箭头往下可到达
CPT	条件概率表 $P(X \mid parents)$
joint distribution	所有变量联合概率
factorization	把 joint 拆成 CPT 乘积
d-separated	图保证条件独立
active path	没被 evidence 阻断的路径
collider	箭头撞到中间: $\rightarrow B \leftarrow$

24. True/False 答题句库

False: 存在 active path

False. Although the path ... is blocked by the observed node ..., there is another active path Therefore the variables are not d-separated given the evidence.

True: all paths blocked

True. Every undirected path between the two variables is blocked by the evidence: paths through ... are blocked at ..., and paths through ... are blocked at Hence the variables are d-separated and conditionally independent.

Collider opened by descendant

The path is active because the collider ... has a descendant ... in the evidence.

Non-collider blocked

The path is blocked because ... is a non-collider on this path and is observed.

25. Tutorial D-Separation 最短背诵版

Claim	Ans	Key reason
$Pa \perp\!\!\!\perp Su \mid HW$	F	$Pa \rightarrow HW \leftarrow S \rightarrow Pj \rightarrow Su$ active; HW observed collider
$C \perp\!\!\!\perp H \mid M$	F	$C \rightarrow Pj \rightarrow Su \rightarrow H$ active
$Pa \perp\!\!\!\perp S \mid Su$	F	HW collider has observed descendant Su
$Pa \perp\!\!\!\perp C \mid H$	F	$Pa \rightarrow H \leftarrow M \leftarrow C$ opened
$Pa \perp\!\!\!\perp C \mid Su, Pj, S$	T	all paths blocked; S or Pj blocks, H unobserved collider

26. 最后 60 秒 Quick Lookup

- 1. BN = DAG + CPT.
- 2. Joint = multiply $P(node \mid parents)$ for every node.
- 3. Boolean CPT params = $2^{\#parents}$; full joint = $2^N - 1$.
- 4. $X \perp\!\!\!\perp Y$: $P(x, y) = P(x)P(y)$.
- 5. $X \perp\!\!\!\perp Y \mid Z$: $P(x, y \mid z) = P(x \mid z)P(y \mid z)$.
- 6. Chain/Fork observed middle $\Rightarrow$ block.
- 7. Collider unobserved $\Rightarrow$ block; observed collider/descendant $\Rightarrow$ open.
- 8. All paths blocked $\Rightarrow$ independent; one active path $\Rightarrow$ not independent.

27. 检查清单 Before Submit

- 写 joint 时每个 node 是否刚好出现一次?
- Root node 是否写 $P(X)$, 不是 $P(X \mid \dots)$?
- CPT 参数有没有用 2^k , full joint 有没有减 1?
- D-sep 是否先忽略箭头找 all paths?
- 是否只检查 path 中间节点, 并按这条路径上的箭头分类?
- 是否把 collider 和 non-collider 的 evidence 规则搞反?
- 是否记得 observed descendant 也会打开 collider?
- 是否区分 graph-guaranteed independence 和具体 CPT 巧合独立?
- False 题是否写出了至少一条 active path?
- True 题是否说明 all paths blocked?

最稳策略: 遇到独立性题不要凭直觉。把路径写出来, 逐段标 “chain/fork/collider + observed?”, 最后给 True/False。